

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1)

MA1513

Lecture Problem Set 3

Instructions

- To be discussed during lecture in week 3 (Aug 24 and 25, 2026).
 - Note that you do not need to submit the answers to this problem set.
 - To prepare for the problem set, students should view the "Week 3 key video" before the lecture.
 - Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.
1. Find a basis for and the dimension of the solution space of each of the following homogeneous systems.

$$(a) \quad \begin{cases} x_1 + 3x_2 - x_3 + 2x_4 = 0 \\ -3x_2 + x_3 = 0 \end{cases} \quad (b) \quad \begin{cases} x_1 + 3x_2 - x_3 + 2x_4 = 0 \\ -3x_2 + x_3 = 0 \\ x_1 - x_4 = 0. \end{cases}$$

2. The regular coordinate axes for the xyz -space is rotated about the z -axis through 45° counterclockwise, viewing from the positive z -axis.

(a) Find a basis S consisting of unit vectors that determine the new coordinate axes.

(b) What are the coordinates of the vector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ relative to the new axes?

(c) Find a matrix $\mathbf{M}$ such that $\mathbf{M}\mathbf{v} = (\mathbf{v})_S$.

$$3. \text{ Let } \mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

(a) Show that the linear system $\mathbf{Ax} = \mathbf{b}$ has no solution.

(b) Find the least squares solution to $\mathbf{Ax} = \mathbf{b}$.

(c) Find the projection of $\mathbf{b}$ onto $\text{span}\{(1, 0, -1, 1), (-1, 1, 0, 1), (0, -1, 1, 1)\}$.

(d) Use (c) to find a vector that is orthogonal to all the three vectors:
 $(1, 0, -1, 1), (-1, 1, 0, 1), (0, -1, 1, 1)$.

4. Given two 3×4 matrices $\mathbf{A}$ and $\mathbf{B}$ with respective row echelon forms

$$\mathbf{R}_A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \mathbf{R}_B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Do we have enough information to find the following? Justify your answers.

- (i) Matrices $\mathbf{A}$ and $\mathbf{B}$.
- (ii) The row spaces of $\mathbf{A}$ and $\mathbf{B}$.
- (iii) The column spaces of $\mathbf{A}$ and $\mathbf{B}$.

Write down a basis for each of the row spaces and column spaces of $\mathbf{A}$ and $\mathbf{B}$ if possible.

Hints

1. First find the general solution of the system and rewrite it as a linear combination of some linearly independent vectors. (a) $\{(0, \frac{1}{3}, 1, 0), (-2, 0, 0, 1)\}$.
2. (a) Locate the positions of the standard basis vectors $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ after the rotation. $\left\{(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0), (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0), (0, 0, 1)\right\}$. (b) $(\sqrt{2}, 0, 1)$. (c) You need to convert certain vector equation into matrix equation form. $\mathbf{M} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1}$
3. (b) Solve the system $\mathbf{A}^T \mathbf{A} \mathbf{x} = \mathbf{A}^T \mathbf{b}$. We get $x = \frac{1}{3}, y = \frac{1}{3}, z = \frac{1}{3}$.
 (c) How is the linear span related to the matrix $\mathbf{A}$? Ans: $(0, 0, 0, 1)$.
 (d) Refer to Chapter 2 notes page 31.
4. (i) No for both; (ii) Yes for both; (iii) Yes for $\mathbf{B}$.